

EVSP-DR

Research update

Week of 21 September 2026
Integer routes, charging constraints, and next tests

What changed this week

Eight jobs finished; completion is not the same as convergence.

New pricing recovered 8 buses in 3 of 4 fresh $k=8$ cases.

Chain 1 also reaches 8 when its dive incumbent is passed to MIP.

Capacity pricing improved; $k=2$ and $k=3$ still hit the time limit.

The strict large case spent its budget building the graph.

Four fresh pools exclude an 8-bus solution

Chain	LP weight	Pool MIP	Added*	After*
C1	8	9	8	8
C3	8	9	8	8
C4	8	9	8	8
C5	8	9	8	8

* Add known sequential routes to the fresh pool, then re-solve.
Both MIP columns show proved fleet minima within their pools.

Read the proof in Gurobi's log

Chain 1, k=8: stage 1 minimizes only the number of buses.

194 trip rows • 39,940 route variables

“Optimal solution found (tolerance 1.00e-04)”

“Best objective 9 ... best bound 9 ... gap 0.0000%”

Incumbent 9 + bound 9 $\Rightarrow$ no 8-bus cover in this pool.

Adding eight witness routes: objective 8, bound 8.

The later charging-stage gap does not undo this fleet proof.

Why a good LP can give a poor integer pool

The LP spreads 8 units of route weight over 80–101 routes.

Whole buses need a compatible set of complete routes.

LP pricing looks for improvement in the fractional objective.

A route can cost slightly more fractionally, yet complete a better integer cover. Its reduced cost can be positive.

Sequential runs retain routes from earlier dual prices.

Diving changes the dual prices by fixing part of a cover.

A small example: fractional value versus compatibility

Six trips: A, B, C and D, E, F. Each route below costs 100.

Pool: AB, AC, BC, DE, DF, EF.

LP: take half of every route $\rightarrow$ 3 route units, cost 300.

Integer pool: two routes per triangle $\rightarrow$ 4 buses, cost 400.

Add AD at cost 101. Reduced cost = $101 - 50 - 50 = +1$.

Now AD + BC + EF covers all trips with 3 buses, cost 301.

The better integer cover need not improve the LP.

Can new pricing recover eight buses?

Chain	Controls	Dive + MIP	Follow-up
C1	9	9	8
C3	9	8	—
C4	9	8	—
C5	9	8	—

Original pilot: 3/4 successes. C1 follow-up: +29.6 minutes.

Same distinct routes; passing the dive incumbent recovered eight.

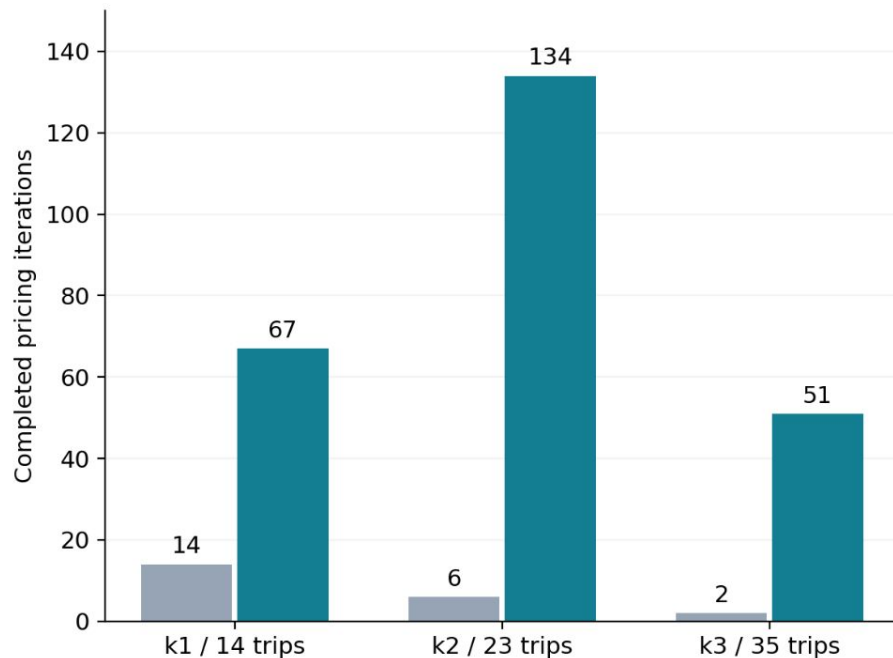
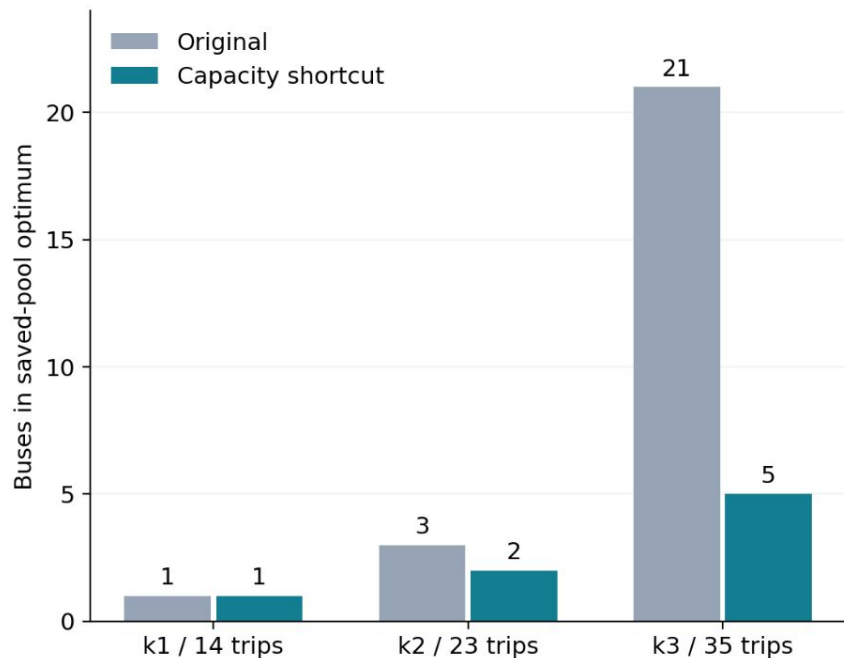
Twelve 12-hour searches still miss 15

Chain	Plain MIP	Heuristics	Fleet bound
C1	18	18	15
C2	17	17	15
C3	17	16	15
C4	19	17	15
C5	16	16	15
C6	18	17	15

All searches timed out; none proved 15 impossible.

The charging stages have also finished. “Heuristics” changes MIP search settings.

Capacity pricing: progress, but not yet convergence



4 h budgets. Only k=1 with the shortcut certified CG (26.9 min); the other five timed out.

The strict large run stalled before pricing

k=15: 4 h 17 min to build 260.7 million arcs; zero pricing.

k=16: six-hour timeout, peak memory 211.9 GiB.

Fix: packed arcs and deferred tie-break keys; physics unchanged.

44 tests pass, plus five fixed-dual and physical replay checks.

26-trip audit: graph build 121 $\rightarrow$ 42 s; memory 2.06 $\rightarrow$ 0.147 GiB.

The new k=16 CG is running; its large-case result is pending.

Remove duplicate service, then optimize charging

Choose exactly one bus to serve each duplicated trip.

Retain movement if needed; waiting at the wrong stop is invalid.

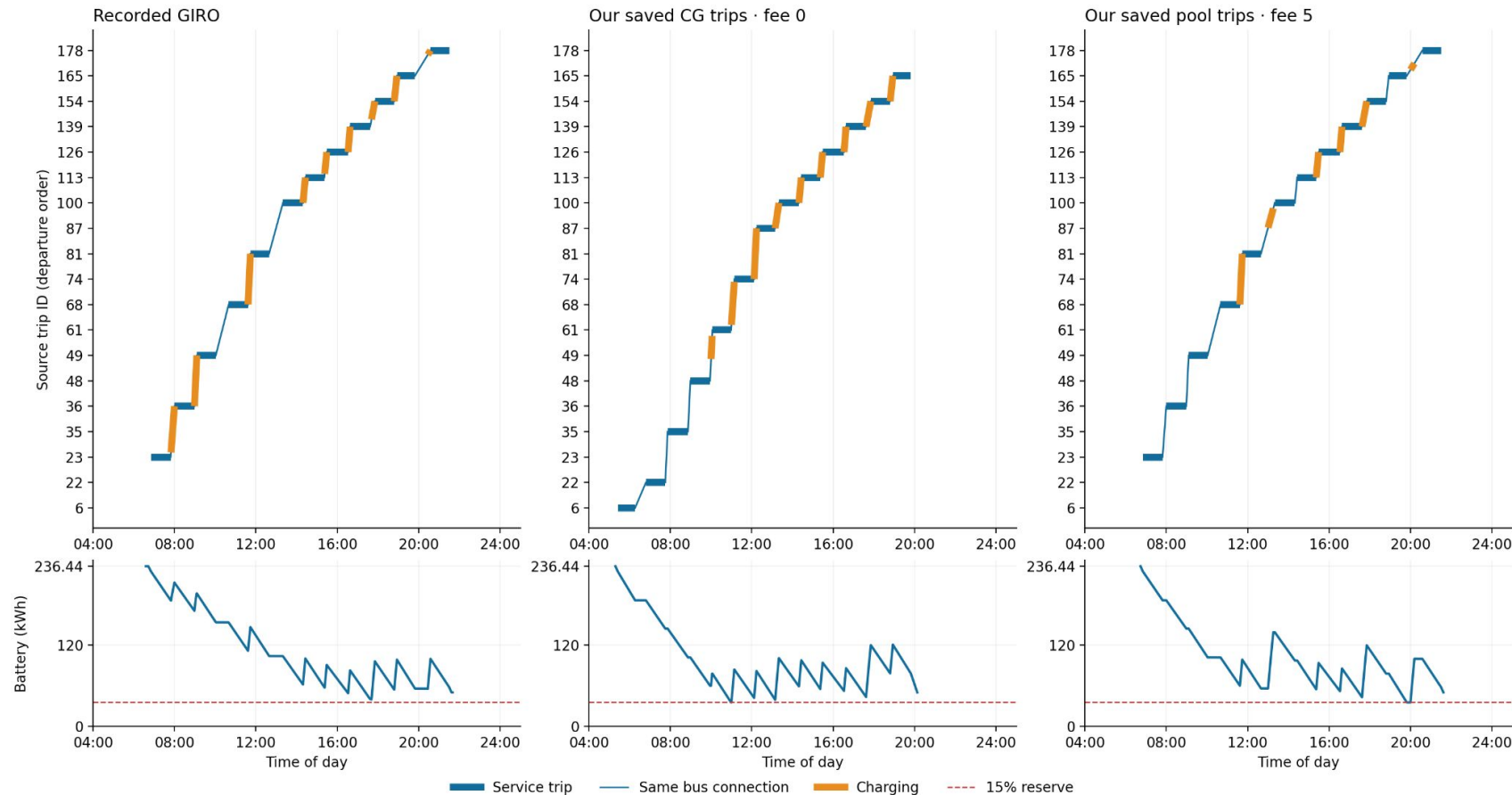
Try shorter connections, then check time, SOC and chargers.

Reoptimize charging after making the service assignment.

Tested $k=5$: 5 buses, 62 trips served exactly once.

Charging energy fell 80.87 kWh; electricity cost fell 2.58.

k=5: charging under matched vehicle physics



What is matched—and what this test measures

236.44 kWh battery • 15% reserve • PARX at 60 kW

Documented opportunity-charge taper • minimum 3 minutes

One charger per opportunity site • equal terminal energy

Five buses and the same 62 trips, served exactly once

Saved trip sequences recharged; no fresh CG under this physics.

Static deadhead; platform/FIFO and crew rules remain unchecked.

Capacity checked for these five buses, without the other duties.

Matched $k=5$: fewer starts with a start fee

Schedule	Electricity	Starts	Start fees	Total
GIRO, fee 0	230.64	52	0	230.64
Ours, fee 0	158.71	42	0	158.71
GIRO, fee 5	230.64	52	260	490.64
Ours, fee 5	183.71	30	150	333.71

Tariff cost units; five-bus totals. Same terminal energy.

Original charging windows leave no material fixed-duty saving.

New sequences: post-hoc charging optimization, not a fresh CG test.